MSG No. 84.53 $P4'_2/m$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,7,8]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,5,6]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[3,4,5,6]`

Table 7: Wyckoff site: `4g`, site symmetry: `2..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2]`
2	$[0, 0, -z]$	`[3,4]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	[5,6]
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	[7,8]

Table 8: Wyckoff site: 4h, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	[1,2]
2	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	[3,4]
3	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[5,6]
4	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[7,8]

Table 9: Wyckoff site: 4i, site symmetry: 2..

No.	position	mapping
1	$\left[0,\, \frac{1}{2},\, z\right]$	[1,2]
2	$\left[0,\, \frac{1}{2},\, -z\right]$	[3,4]
3	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	[5,6]
4	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-z\right]$	[7,8]

Table 10: Wyckoff site: 4j, site symmetry: m..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,4]
2	$[-x,\, -y,\, 0]$	[2,3]
3	$\left[-y,\, x,\, \frac{1}{2}\right]$	[5,8]
4	$\left[y,\, -x,\, \frac{1}{2}\right]$	[6,7]

Table 11: Wyckoff site: 8k, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-x,\, -y,\, -z]$	[3]
4	$[x,\, y,\, -z]$	[4]
5	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[5]
6	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[6]
7	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[7]
8	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[8]